Vadim Prokhorov

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Paris region, France

PROFILE

Looking for a summer internship, July to August 2027, in data science, machine learning or data engineering. First-year engineering student at **Télécom Paris** (Institut Polytechnique de Paris), admitted through the GEI-UNIV competitive exam (20 places on this track), after a BSc in Computer Science with a Mathematics minor at **Sorbonne Université** (**ranked 2nd of 57** in year 2). **Four data roles** completed, including a model and a service shipped to production at DOCENT. I work on the mathematical side of data science: hypothesis testing, evaluation metric design, Bayesian modelling. And I put what I design into production (Python, FastAPI, PostgreSQL, Docker, AWS).

EDUCATION

Engineering degree (Diplôme d'ingénieur), year 1 Télécom Paris, Institut Polytechnique de Paris, Palaiseau — 2026 - 2029

Admitted through GEI-UNIV, a highly selective competitive exam: 20 places at Télécom Paris on this track. Expected graduation 2029. Year-1 coursework: probability, functional and Fourier analysis, digital communications and information theory, logic.

BSc Computer Science, Mathematics minor Sorbonne Université, Faculty of Science and Engineering, Paris — 2023 - 2026

Yearly averages: year 1 **17.49/20** | year 2 **15.92/20, ranked 2nd of 57** | year 3 **17.53/20**. Data Science 95.75/100 (**1st of 117**) | Networks 100/100 (**1st of 316**) | Algorithms II (3rd of 318) | Data Structures (4th of 392) | Algorithms I (6th of 496) | Mathematics for Science 1 and 2: 96.6/100 and 98/100 (36th of 2227).

EXPERIENCE

DOCENT — Mar - Aug 2026

Data Scientist Intern

Cultural trend analysis and prediction platform | part-time, then full-time | English-speaking team

- **Segmentation method put into production**, selected from 15 algorithm configurations evaluated on 8 independent tests, using a reproducible benchmark that replays the whole study in 2 minutes (scikit-learn, scipy, UMAP). Deterministic segmentation, macro-F1 **0.776**.
- **Custom metric:** an overlap score with an analytical ceiling, built because degenerate partitions kept winning the standard metrics.
- **Stopped tuning on statistical evidence:** three independent tests (Hopkins statistic between 0.54 and 0.60, co-association between methods, inter-annotator agreement) showed the data had no natural cluster structure. Two experts agreed with each other less (ARI 0.170) than the chosen model agreed with them (ARI 0.189): the ceiling was the human reference.
- **Caught a measurement artifact before publication:** on an identical partition, the silhouette score rose from 0.018 in the original space to 0.326 after UMAP projection. Decision: cluster in the original space, keep projections for visualisation.
- **Brand-to-artist matching engine** raised to **0.768** top-1 precision against 0.140 for a random draw, comparing 18 methods under three evaluation protocols (stratified hold-out, real cases, bootstrap); Bayesian Bernoulli-Beta variant for uses that need a calibrated probability.
- **LLM annotation pipeline industrialised:** three no-code workflows (n8n) replaced by a production Python service (FastAPI, PostgreSQL, ~60 REST endpoints, Docker, AWS), with per-run cost tracking and recovery from partial failure. Platform running on ~1,000 artists, designed for tens of thousands (indexed PostgreSQL, worker containers that scale with load, asynchronous processing).

Agency 1301 — Autumn 2025

Data Scientist (freelance)

Detection of unauthorised republication of photographs (copyright)

- **Made the detection rate interpretable:** typed failure statuses separate a score that cannot be computed from a score that is genuinely zero, so a missing result is no longer confused with a technical failure, which removed a silent bias from the statistics.
- **Made the decision threshold re-examinable:** every candidate persisted, rejected as well as retained (100 to 200 per photo), so recalibrating no longer means rerunning searches that took several hours per batch. Threshold, version and timestamp frozen on every run; ground truth annotated on 6 levels.
- Full PostgreSQL database sized for their data volumes, opened to non-technical users through NocoDB with access control.

Prana France

May - Aug 2025

Data Analyst Intern

Industrial SME, heat-recovery and decentralised ventilation

- Power BI dashboards on sales, commercial activity and lead generation, built on data prepared in SQL.
- Customer-support **RAG** assistant built in Python on embeddings of the product documentation, deployed on AWS.

SAS Datafoncier

Jun - Aug 2024

Data Scientist Intern

Start-up, real-estate market analysis

- Real-estate market prediction by regression (price, surface area, trends): hyperparameter tuning, cross-validation, class rebalancing, missing data; **ResNet** feature extraction on property images to estimate their condition.

SKILLS

Mathematics & statistics: Hypothesis testing | bootstrap | agreement metrics (ARI, Cronbach's alpha) | Hopkins | Bayesian inference (Bernoulli-Beta, conjugate priors, shrinkage, calibration) | PCA, Procrustes analysis | metric design and correction for chance

Machine learning: scikit-learn | PyTorch | TensorFlow | scipy | classification & regression | clustering (KMeans, Ward, HDBSCAN) | dimensionality reduction (PCA, UMAP, t-SNE, MDS) | evaluation protocol design

Generative AI & LLMs: RAG | embeddings | vector databases | prompt engineering | agentic systems | sentence-transformers | CLIP | n8n

Data & backend: Python | C | C++ | SQL | PostgreSQL (schema design, indexing, scaling) | FastAPI | Pandas | NumPy | NocoDB | Power BI

Infrastructure: Docker | AWS (EC2, RDS, S3) | asynchronous and parallel processing | Git | Linux | pytest | Claude Code, Cursor, MCP

Languages: Russian (native) | French (C1) | English (B2, TOEFL iBT 87/120, daily working language at DOCENT) | Driving licence B

Interests: Tennis (strong level) | chess | competitive programming | Kaggle